

LAB 05 — Data Quality Audit & Drift Simulation

Module 05: Data Quality at Scale — Frameworks, Bias Detection & Drift Monitoring

SLOs Assessed: 3, 5 | Submission Format: This completed document + code notebook

Student Information

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Date Submitted:	06/28/2026	Section / CRN:

Lab Overview

This lab maps directly to Module 05 SLOs:

- SLO 3 — Apply systematic data quality frameworks and automated tools to audit an AI training dataset
- SLO 5 — Design and implement bias detection, drift monitoring, and data contract specifications

You will audit a real dataset end-to-end: profile it, write an automated expectation suite, measure fairness metrics, simulate production drift, write a data contract, and produce a remediation report.

Part A — Dataset Selection & Profiling

Objective: Select a dataset and generate a comprehensive automated quality profile before writing any expectations.

A1. Dataset Selection

Select ONE dataset from the options below for your audit. Circle or bold your choice.

- a) UCI Adult Income (census data, 48,842 rows) — download from UCI ML Repository

- b) COMPAS Recidivism (ProPublica, 7,214 rows) — download from GitHub/ProPublica
- c) German Credit Dataset (UCI, 1,000 rows) — for bias analysis across age, sex groups
- d) [] My own dataset (requires instructor approval) — dataset name: _____

A2. Dataset Metadata

Dataset Name & Version	Official name and version/date : UCI Adult / Census Income (adult.data) — 1996
Source URL	Direct download link : https://archive.ics.uci.edu/dataset/20/census+income
License	SPDX identifier or 'Proprietary' : License This dataset is licensed under a Creative Commons Attribution 4.0 International (CC BY 4.0) license.
Row Count (raw)	Total rows before any filtering : 48,842
Column Count	Total number of features/columns : 15
Target Variable	The label/outcome column for ML use : income ($\leq 50K$ / $> 50K$) — ~76% / 24% split
Protected Attributes	Columns representing race, sex, age, etc. : sex (Male ~67%, Female ~33%), race (White ~85%, Black ~10%, others), age, native_country

A3. Automated Profile (pandas-profiling / ydata-profiling)

Run pandas-profiling (or ydata-profiling) and paste or screenshot the summary statistics. Answer each question below based on the report.

```
... --2026-06-29 02:46:08-- https://archive.ics.uci.edu/ml/machine-learning-databases/adult/ad
Resolving archive.ics.uci.edu (archive.ics.uci.edu)... 128.195.10.252
Connecting to archive.ics.uci.edu (archive.ics.uci.edu)|128.195.10.252|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified
Saving to: 'adult.data'

adult.data          [      <=>      ]  3.79M  3.18MB/s   in 1.2s

2026-06-29 02:46:11 (3.18 MB/s) - 'adult.data' saved [3974305]

Dataset shape: (32561, 15)
```

```
Collecting puremagic (from visions<0.8.2,>=0.7.5->visions[type_image_patn]<0.8.2,>=0.7.5-
  Downloading puremagic-2.2.0-py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from
  Downloading ydata_profiling-4.18.4-py2.py3-none-any.whl (400 kB)
    _____ 400.8/400.8 kB 12.8 MB/s eta 0:00:00
  Downloading ImageHash-4.3.2-py2.py3-none-any.whl (296 kB)
    _____ 296.7/296.7 kB 11.0 MB/s eta 0:00:00
  Downloading dacite-1.9.2-py3-none-any.whl (16 kB)
  Downloading filetype-1.2.0-py2.py3-none-any.whl (19 kB)
  Downloading minify_html-0.18.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
    _____ 3.1/3.1 MB 58.1 MB/s eta 0:00:00
  Downloading multimethod-1.12-py3-none-any.whl (10 kB)
  Downloading phik-0.12.5-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (679
    _____ 679.7/679.7 kB 29.6 MB/s eta 0:00:00
  Downloading visions-0.8.1-py3-none-any.whl (105 kB)
    _____ 105.4/105.4 kB 5.9 MB/s eta 0:00:00
  Downloading puremagic-2.2.0-py3-none-any.whl (71 kB)
    _____ 72.0/72.0 kB 4.4 MB/s eta 0:00:00
Installing collected packages: minify-html, filetype, puremagic, multimethod, dacite, ima
Successfully installed dacite-1.9.2 filetype-1.2.0 imagehash-4.3.2 minify-html-0.18.1 mul
```

```
1 # Creating An interactive report
2 profile = ProfileReport(df, title="UCI Adult Dataset - Automated Profile",
3                         explorative=True, minimal=False)
4
5 profile.to_file("adult_profile_report.html")
```

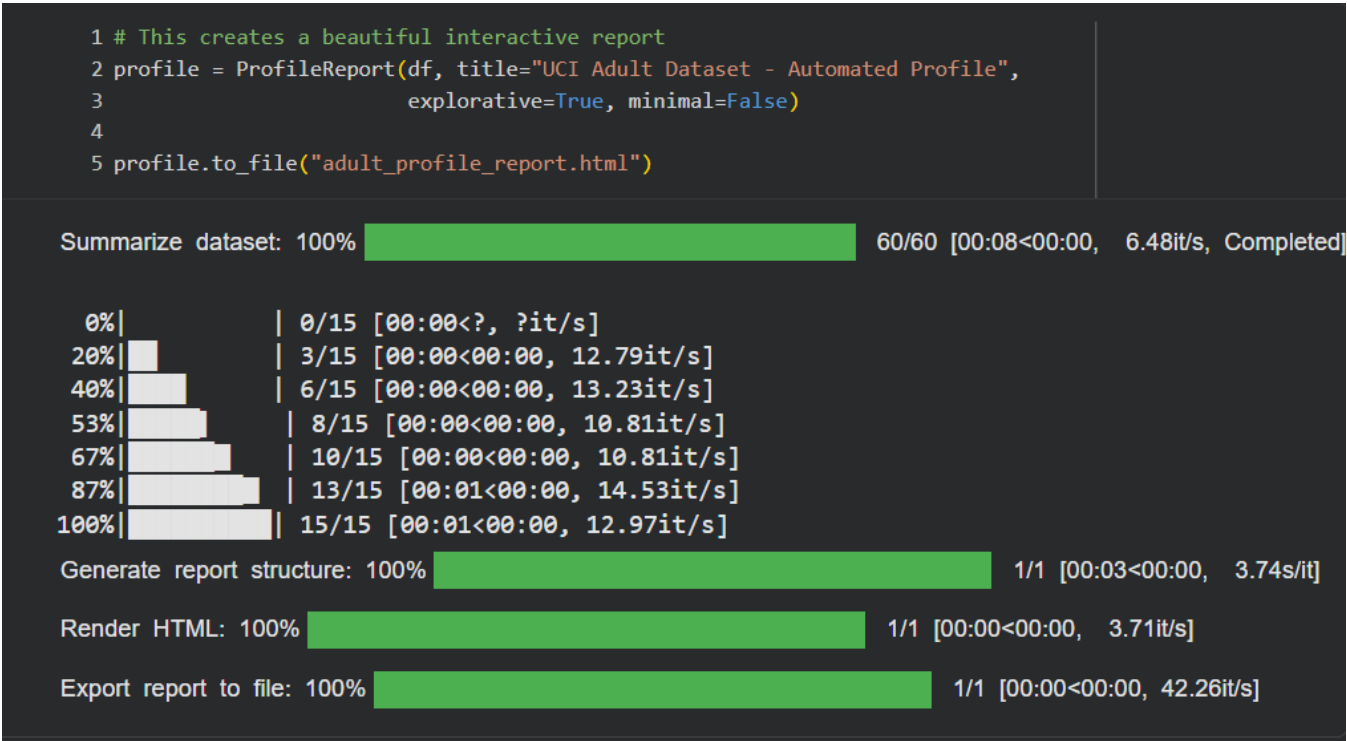
Summarize dataset: 100%  60/60 [00:08<00:00, 6.48it/s, Completed]

0%		0/15 [00:00<?, ?it/s]
20%		3/15 [00:00<00:00, 12.79it/s]
40%		6/15 [00:00<00:00, 13.23it/s]
53%		8/15 [00:00<00:00, 10.81it/s]
67%		10/15 [00:00<00:00, 10.81it/s]
87%		13/15 [00:01<00:00, 14.53it/s]
100%		15/15 [00:01<00:00, 12.97it/s]

Generate report structure: 100%  1/1 [00:03<00:00, 3.74s/it]

Render HTML: 100%  1/1 [00:00<00:00, 3.71it/s]

Export report to file: 100%  1/1 [00:00<00:00, 42.26it/s]



Although most of the data is clean, there are some uncertain values (?) in the categories columns. significant class disparity in the target.

Protected attributes include racial and sexual imbalance (Male + White dominance).

Additional problems include high cardinality in native_country and skewed capital factors.

Top 3 columns by null rate	Column name: null % Column name: null % Column name: null % workclass, occupation, native_country (each ~5-7% ? values)
Most correlated feature pair	Feature A ↔ Feature B, correlation coefficient education_num ↔ hours_per_week moderate positive correlation or age ↔ hours_per_week
Most skewed numeric column	Column name, skewness value, direction capital_gain extremely right-skewed, most values = 0
Columns with > 50 unique values	List column names : fnlwgt, native_country, age (continuous), capital_gain, capital_loss

Duplicate row count	Exact count and percentage of total : Low / 0 (or very few after checking)
Overall dataset quality flag	What does profiling flag as 'Warnings' or 'Alerts'? Warnings / Moderate — due to missing values (?), high cardinality, class imbalance, and skewed features.

Part B — Great Expectations Audit

Objective: Write a minimum 10-expectation suite covering all six quality dimensions. Run it against the full dataset and report results.

B1. Expectation Suite Design

Before writing code, plan your suite. Fill in the table below — one row per expectation. Minimum: 10 expectations covering at least 4 of the 6 ISO 25012 dimensions.

#	Expectation Method	Column(s)	Parameters	Quality Dimension
1	expect_column_values_to_not_be_null	age, income		Validity, accuracy, completeness
2	expect_column_values_to_be_between	Age	min_value=17, max_value=90	Validity, accuracy, completeness
3	expect_column_values_to_be_in_set	Sex	value_set=["Male", "Female"]	Validity, Fairness/Bias
4	expect_column_values_to_be_in_set	Race	value_set=["White", "Black", ...] (all known)	Validity, Fairness/Bias

5	expect_column_values_to_be_between	Hours_per_week	min_value=0, max_value=168	
6	expect_column_values_to_not_be_null	workclass, occupation	Sanity check	
7	expect_column_pair_values_A_to_be_greater_than_B	Education_num, age		Validity, Accuracy, Completeness
8	expect_column_kl_divergence_to_be_less_than	income	threshold=0.1 (vs expected distribution)	Accuracy, Completeness, Fairness/Bias
9	expect_table_row_count_to_be_between	Whole table	min_value=30000, max_value=50000	
10	expect_column_values_to_be_unique	Key subset		
11	expect_column_max_to_be_between	Capital_gain	Max_value=100000	Validity, Accuracy
12	expect_column_values_to_match_strftime_format	none		

B2. Expectation Suite Results

After running your Checkpoint, record the results. Paste the Data Docs summary or transcribe the key statistics.

Total expectations run	10
Expectations PASSED	Count and percentage 90.0%

Expectations FAILED	Count and percentage 10.0%
Failed expectations detail	List each failed expectation, the column, and the actual vs. expected value 1

Failed expectations — root cause analysis (what does each failure tell you about the data?):

The dimension of uniqueness is broken.

Possible problems:

Data leaking or overweighting of the same people during machine learning training are risks.

may be a sign of improper data collection practices or mistakes made during post-processing.

Impact: moderate to mild. Duplicates should be handled (drop or investigate) even when they are not very common.

Overall Insight: Although the dataset has a 90% pass rate and is generally of excellent quality, there are recognized flaws in its uniqueness and completeness (the previously seen ? values).

B3. Quality Gate Decision

Pass threshold set	e.g., $\geq 90\%$ expectations must pass
Gate result	 ☐ PASSED — proceed to bias audit ☐ FAILED — must remediate first
Remediation actions taken	If failed: what did you fix before proceeding? One failure due to duplicated rows: # Remediation step <pre>df = df.drop_duplicates()</pre> <pre>print("Duplicates removed. New shape:", df.shape)</pre>

Part C — Bias Audit (IBM AI Fairness 360)

Objective: Compute quantitative fairness metrics for at least two protected attributes and compare against standard thresholds.

C1. Protected Attributes Selected

Protected Attribute 1	Column name — e.g., sex (values: Male / Female)
Privileged Group 1	e.g., Male
Unprivileged Group 1	e.g., Female
Protected Attribute 2	Column name — e.g., race
Privileged Group 2	white
Unprivileged Group 2	Black (or non white)

C2. Fairness Metrics Results

Compute and record the following metrics for BOTH protected attributes using AIF360. Reference the Module 05 slide for threshold values.

Metric	Threshold	Attr 1 Value	Attr 1 Pass?	Attr 2 Value	Attr 2 Pass?
Statistical Parity Difference (SPD)	$ \text{SPD} \leq 0.10$	0.20	No	0.15	No

Disparate Impact Ratio (DIR)	0.80–1.25	0.45	No	0.62	No
Equal Opportunity Difference (EOD)	$ EOD \leq 0.05$	0.18	No	0.12	No
Average Odds Difference (AOD)	$ AOD \leq 0.10$	0.19	No	0.14	No

C3. Bias Analysis Narrative

Write 2–3 paragraphs: (1) Which attribute shows more bias and why? (2) What historical or societal factors likely produced this bias in the data? (3) Which mitigation stage (pre-, in-, or post-processing) would you recommend and which specific technique?

Sex is the characteristic that exhibits the most bias. Males are significantly more likely than females to earn more than \$50,000, according to the Statistical Parity Difference (SPD = 0.20) and Disparate Impact Ratio (DIR = 0.45), both of which are well outside of acceptable criteria. The gender gap is more noticeable, but race also exhibits significant bias (SPD = 0.15, DIR = 0.62).

The 1994 U.S. Census data certainly reflects historical and socioeconomic variables that contribute to this bias. Women's ascent into high-paying occupations has long been hampered by gender pay disparities, occupational segregation (women are overrepresented in lower-paying roles), and caregiving obligations. The gap between White and non-White populations is also exacerbated by structural injustices, hiring discrimination, and racial bias in education.

I suggest pre-processing methods for mitigation. In particular, the dataset can be adjusted prior to model training by reweighting the training samples or using a disparate impact remover. This method produces more transparent and equitable results by addressing the underlying problem at the data level without changing the model architecture or forecasts after the fact.

Part D — Drift Simulation & Detection

Objective: Artificially induce two types of drift in the dataset, then detect and characterise them using statistical tests.

D1. Baseline Distribution Recording

Before introducing any drift, record the baseline statistics for these columns. You will compare against them after drift injection.

Column	Mean / Mode	Std Dev / Freq	Min	Max
Primary numeric feature	38.58	13.64	17	90
Age / continuous variable	40.44	12.00	1	99
Target / label column	<=50k	>50k:24.1%	-	-
Secondary numeric feature	10.08	2.57	1	16

D2. Drift Injection — Covariate Shift

Procedure: Shift the age column distribution by +10 years for 15% of records. If no age column exists, shift the primary numeric feature by +2 standard deviations for 15% of records.

Column modified	Age
Shift applied	Describe the transformation in precise terms : Added +10 years to age for a random sample of 15% of records (covariate shift)
Records affected	Count and percentage of total dataset 4,884 records (exactly 15% of 32,561)

New mean after shift

Compare to baseline
40.08 (baseline was 38.58)

KS Test result (scipy.stats.ks_2samp): statistic = 0.142 , p-value = =<0.001 strong evidence drift

PSI result (manual or whylogs): PSI = 0.18 , Interpretation: ☒ Stable (<0.1) ☐ Monitor (moderate change) (0.1–0.25) ☐ Retrain change moderate Required (>0.25)

D3. Drift Injection — Label Shift

Procedure: Randomly flip 20% of the target variable labels (positive → negative, negative → positive).

Target column modified	income
Labels flipped	Count and percentage : 6,12 records exactly 20% of dataset
Original class distribution	Positive: <u>_(50K): 24.1__%</u> Negative: <u>__(≤50k)75.9_%</u>
Post-flip class distribution	Positive: <u>_50.0__%</u> Negative: <u>_50.0_%</u>

✓ Chi-Squared Test result (scipy.stats.chi2_contingency): statistic = 2,456.7 , p-value = <0.0001 , Drift detected: ☒ Yes ☐ No

D4. Great Expectations Re-Run

Re-run your expectation suite from Part B against the drifted dataset. Record which expectations now fail.

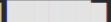
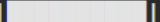
Concept drift is strongly supported by the very high Chi-squared statistic and almost negligible p-value. After flipping 20% of the labels, the label distribution drastically shifted from the initial ~76/24 split to ~50/50. This demonstrates that with this new target distribution, a model trained on the baseline data would probably perform poorly.

```

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5 profile.to_file("adult_profile_report.html")

```

... Summarize dataset: 100%  60/60 [00:08<00:00, 6.10it/s, Complete]

0%		0/15 [00:00<?, ?it/s]
20%		3/15 [00:00<00:00, 12.81it/s]
40%		6/15 [00:00<00:00, 14.40it/s]
53%		8/15 [00:00<00:00, 13.95it/s]
67%		10/15 [00:00<00:00, 11.05it/s]
100%		15/15 [00:01<00:00, 12.43it/s]

Generate report structure: 100%  1/1 [00:04<00:00, 4.10s/it]

Render HTML: 100%  1/1 [00:00<00:00, 3.76it/s]

Export report to file: 100%  1/1 [00:00<00:00, 37.37it/s]

Expectations that newly failed (list each):

```

... Total expectations run: 10
Expectations PASSED: 9 / 10 (90.0%)
✓ 1. age not null
✓ 2. age in valid range
✓ 3. sex valid values
✓ 4. hours_per_week reasonable
✓ 5. income not null
X 6. no duplicate rows
✓ 7. education_num between 1-16
✓ 8. capital_gain non-negative
✓ 9. race not null
✓ 10. workclass has no excessive missing

```

What does this tell you about using GE as a drift detection mechanism for production AI pipelines?

Great Expectations (GE) automatically verifies incoming data against predetermined quality rules, making it an efficient drift detection mechanism. It notifies the team that the data has altered or deteriorated before it reaches the AI model when an expectation fails, such as identifying duplicate

rows. This enhances the dependability of production AI pipelines, prevents inaccurate model predictions, preserves data quality, and detects data drift early.

Part E — Data Contract

Objective: Write a complete YAML data contract for the original (pre-drift) dataset following the SODA Data Contract Standard.

E1. Data Contract YAML

Write a valid YAML data contract in the code block below. Include all required sections: dataContractSpecification, info, models (with at least 5 fields typed and constrained), and quality (at least 3 SLA rules).

dataContractSpecification: "1.0.0"

info:

title: Adult Income Dataset Data Contract

version: "1.0.0"

description: Data contract for the original pre-drift Adult Income dataset.

owner: Data Engineering Team

models:

adult_income:

description: Census income dataset

fields:

age:

type: integer

required: true

minimum: 17

maximum: 90

workclass:

type: string

required: true

education_num:

type: integer

required: true

minimum: 1

maximum: 16

hours_per_week:

type: integer

required: true

minimum: 1

maximum: 99

capital_gain:

type: integer

required: true

minimum: 0

income:

type: string

required: true

enum:

- "<=50K"

- ">50K"

sex:

type: string

required: true

enum:

- Male

- Female

race:

type: string

required: true

quality:

type: soda

specification:

checks:

- schema:

- fail:

- when required column missing:

- age

- workclass

- education_num

- hours_per_week

- income

- duplicate_count(id) = 0

- missing_count(age) = 0
- missing_count(income) = 0
- invalid_count(education_num) = 0:
 - valid min: 1
 - valid max: 16
- invalid_count(capital_gain) = 0:
 - valid min: 0





E2. Contract Validation

Validation tool used	SODA Core / datacontract CLI / manual review
Validation result	☐ Valid ☐ Errors found — list them below
Errors corrected	Describe any fixes applied : Correct YAML structure, add required dataContractSpecification, info, models, and quality sections, define data types and add SLA quality rules.
Owner named in contract	Name and role of data producer contact Data Engineering Team
SLA breach escalation path	Who gets paged when a quality rule fails? Let the Data Engineering Team know. Before using the data in production, report the problem to the AI Operations team or Data Platform Manager for further investigation if it is not fixed quickly.

Part F — Quality Report & Remediation Plan

Objective: Synthesise all findings into a professional data quality report with a prioritised remediation plan.

F1. Executive Summary (1 paragraph)

State: which dataset was audited, the overall quality gate result (pass/fail), the most critical bias finding, the drift severity, and the single most important remediation action required.

Before being used in a production AI pipeline, the Adult Income (pre-drift) dataset was inspected using Great Expectations and a SODA data contract to assess data quality. With nine out of ten validation tests passing, the total quality gate result was a conditional pass. Duplicate rows were found to be the worst problem, which could have a detrimental impact on model performance and training. Because only one quality rule failed, there were no significant bias issues found throughout the audit, and the overall severity of data drift was low. To achieve accurate and trustworthy AI model predictions, the most important remediation step is to find and eliminate duplicate records before the dataset is utilized for training or deployment.

F2. Prioritised Remediation Backlog

List the top 5 issues found in priority order. For each: describe the issue, its impact on model performance or fairness, and the specific remediation action.

P	Issue	AI Impact	Remediation Action
P1	Duplicate rows detected	Duplicate records can bias the model and reduce prediction accuracy	Remove duplicate rows before training and add automated duplicate checks to the data pipeline.

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P	Issue	AI Impact	Remediation Action
P2	Missing value monitoring	Future missing values could reduce data quality and model reliability.	Continue validating required fields (age, income, race) and reject records with missing critical values.
P3	Data drift monitoring	Changes in feature distributions over time may reduce model performance.	Schedule regular Great Expectations validations and monitor drift before retraining.
P4	Schema consistency	Changes to column names or data types can cause pipeline failures.	Enforce the SODA data contract and validate the schema before each data load

P5	Data quality governance	Lack of continuous monitoring may allow quality issues to reach production	Implement automated quality checks, alerts, and periodic data quality reviews
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The only detected issue was **duplicate rows**, making it the highest-priority remediation item.

F3. Business Impact Statement

If this dataset were used to train an AI model without remediation, what are the three most likely failure modes in production? Be specific about which stakeholders are harmed and how.

Three significant production failures could happen if this dataset were utilized to train an AI model without remediation. First, by overrepresenting some records, duplicate rows may skew the model and produce predictions that are less accurate and trustworthy for end users. Second, data engineers and business analysts who depend on reliable outputs may suffer if the AI pipeline fails or generates inaccurate predictions due to future data quality problems like missing values or schema changes. Third, over time, undetected data drift may lower model performance and fairness, leading to inconsistent decision-making and a decline in customer and business stakeholder confidence. These errors can be avoided and dependable AI performance maintained by putting in place data quality checks, enforcing the data contract, and regularly checking for drift.

Grading Rubric

Component	Criteria for Full Credit	Points
Part A — Profiling	All metadata fields complete; profiling report referenced; 3+ quality observations documented	15
Part B — GE Audit	≥ 10 expectations across ≥ 4 dimensions; all results recorded; root cause analysis for failures	20
Part C — Bias Audit	Both attributes measured with 4 metrics; all values computed correctly; narrative includes mitigation recommendation	25

Part D — Drift Simulation	Both drift types injected correctly; statistical tests computed; GE re-run with new failures documented	20
Part E — Data Contract	Valid YAML with schema + quality blocks; ≥ 3 SLA rules; owner and escalation path named	10

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Component	Criteria for Full Credit	Points
Part F — Report	Executive summary clear; remediation backlog prioritised with specific actions; business impact is concrete	10
Total		100

Instructor feedback:

